

Empirical Proof Record: REFUTED

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Proof ID: 2df833267cdf3fb1eda61bf8aef0e41a...

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1. Claim

Under sustained load, the substrate stays cognitively alive: (for every cycle in the trajectory, $\Phi \geq 0.05$). The reported floor is the lowest- Φ cycle across the whole run.

- **Operationalisation:** Stream 3 passes over 8 substantive stimuli through Kimera’s entity target (Takwin); read the per-cycle “phi”: `phi_floor_observed` = $\min \Phi$ over all cycles; the LTL invariant holds iff $\min \Phi \geq \varphi_floor$.
- **Threshold:** `phi_floor >= 0.05 phi`
- **H0:** H0: $\min \Phi < 0.05$ — Φ collapses below the non-collapse floor somewhere in the run
- **H1:** H1: $\min \Phi \geq 0.05$ — Φ stays above the collapse floor across the whole trajectory (substrate stays cognitively alive under sustained load)

2. Verdict

REFUTED — observed 0.0

Φ floor 0.0000 over 24 cycles (3 passes $\times$ 8 stimuli); mean Φ 0.4861, band [0.0000, 0.6571], stdev 0.2808; non-collapse rate 75.0%; 0 cycles produced no Φ ; worst cycle: stimulus 1 @ cycle 1 = 0.0000

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.phi_stability	phi_floor	0.0	—	ophamin 0.81.0

4. Pre-registration

- Registered at: 2026-05-21T23:31:15.544136+00:00
- Config hash: fc0d6fa2b433540d0fc987795a9290da...
- Data hash: 3c5602a31f04f6ccaa77bd90fa1416a6...

5. Signature

99f200613aaf5b2667f7dc059a5d6b1eeef253e36d6f392c800ac53deb22e343